

ICS 6B F26 Take Home Exam 2

Due: October 9, 2026 at 11:59PM

Name: Jeremy Zhao

UCI NetID :

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(alpha-numeric; **NOT** your student ID)

- Read the instructions of each question carefully.
- Show your work. Where applicable, partial credit will be given on the work shown.
- Please keep in mind the academic honesty guidelines. This take-home exam is to be completed individually, with no outside help. You may use any resources from our class (ZyBooks and resources from Canvas), but you may not use any other online resources.
- You may choose to print the exam or use a digital editor for completing the exam. It is required that you use this PDF to complete your work. If you have no access to a printer or digital tools to fulfill the exam, feel free to reach out to the staffs regarding your concerns.
- If you have any questions, please post a private Ed or attend available Office Hours. Note that we are not allowed to provide specific help to answering the exam questions.
 - There are **5** problems on this test.
 - There are a total **16** of points.
 - Good luck!



Seat

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Problem 1 (2 points)

Rewrite each of the following statements using logical quantifiers and implications. If you want to indicate x is an integer you can write it as $x \in \mathbb{Z}$. For example, x is an even integer can be expressed using logical quantifiers as $\exists k \in \mathbb{Z} (x = 2k)$.

1. The integer x is a multiple of y .

$$\exists k \in \mathbb{Z} (kx = y)$$

2. The sum of x squared and y squared is equal to square of some integer.

$$\exists k \in \mathbb{Z} (k^2 = x^2 + y^2)$$

3. The product of any two integers is an integer.

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} (xy \in \mathbb{Z})$$

4. The ratio of two integers is not always an integer.

$$\exists x \in \mathbb{Z} \exists y \in \mathbb{Z} \neg (x/y \in \mathbb{Z})$$

Problem 2 (3 points)

Express each of the following predicates and propositions in formal logic notation. The first one has been defined as an example. The domain is all positive integers denoted as $\mathbb{Z}^+$. You may use predicates that are already defined (e.g. $\text{PRIME}(n)$ returns true if n is prime).

$$\text{PRIME}(n) \equiv (n > 1) \wedge \neg(\exists x \exists y : (x > 1) \wedge (y > 1) \wedge (x \cdot y = n))$$

1. $\text{PRIME}^2(n) \equiv n$ is a prime number squared.

$$\exists p : \text{PRIME}(p) \wedge (n = p \cdot p)$$

2. There is a smallest prime number.

$$\exists p : \text{PRIME}(p) \wedge (\forall q : \text{PRIME}(q) \rightarrow p \leq q)$$

3. Every prime number greater than 3 is one less than or one more than a multiple of 6.

$$\forall p : ((\text{PRIME}(p) \wedge (p > 3)) \rightarrow (\exists k : (p = 6k - 1) \vee (p = 6k + 1)))$$

Problem 3 (4 points)

Consider the following predicates over the domain of positive integers

- $A(x)$: x is an even number.
- $B(x)$: x is a multiple of 3.
- $C(x)$: x is a prime number.

Determine the truth value of the following quantified statements.

1. $\exists x (B(x) \wedge C(x))$

☒ True

☐ False

☐ Unknown

2. $\exists x (\neg A(x) \wedge \neg B(x) \wedge \neg C(x))$

☒ True

☐ False

☐ Unknown

3. $\forall x (C(x) \rightarrow (A(x) \wedge B(x)))$

☐ True

☒ False

☐ Unknown

4. $\forall x ((B(x) \wedge C(x)) \rightarrow \neg A(x))$

☒ True

☐ False

☐ Unknown

Problem 4 (2 points)

Simplify the following statements with nested quantifiers until each negation sign applies directly to a predicate (that is, so that no negation is outside a quantifier or an expression involving logical connectives). Writing down the final expressions is sufficient.

For example, $\neg(A(x) \vee B(x))$ is not valid but $\neg A(x) \wedge \neg B(x)$ is valid.

1. $\neg \forall x \exists y (A(y, x) \rightarrow \forall z (B(x) \vee C(y, z)))$

$$\exists x \forall y \exists z (A(y, x) \wedge \neg B(x) \wedge \neg C(y, z))$$

2. $\neg(\exists x \exists y ((B(y) \rightarrow C(x, y)) \vee (\forall z A(z, y))))$

$$\forall x \forall y \exists z (B(y) \wedge \neg C(x, y) \wedge \neg A(z, y))$$

Problem 5 (5 points)

You are a baseball fan. The following predicates are over the domain of professional baseball players.

- $P(x, y)$: x scored as many or fewer runs than y i.e. $\text{runs}(x) \leq \text{runs}(y)$.
- $S(x)$: x is a starting pitcher.
- $D(x)$: x is a designated hitter.
- $L(x)$: x is a player part of a local team.
- $M(x)$: x is an MVP.

Give a logical expression that is equivalent to each English statement.

1. Every player that is a starting pitcher scored (strictly) more runs than some player who is part of a local team.

$$\forall x \exists y (S(x) \rightarrow (L(y) \wedge \neg P(x, y)))$$

2. Every player part of a local team scored as many or fewer runs than some player who is not a starting pitcher.

$$\forall x \exists y (L(x) \rightarrow (\neg S(y) \wedge P(x, y)))$$

3. Every player that is not a starting pitcher and not part of a local team scored as many or fewer runs than Frederick. (Assuming Frederick is a player.)

$$\forall x ((\neg S(x) \wedge \neg L(x)) \rightarrow P(x, \text{Frederick}))$$

4. There is a starting pitcher who scored (strictly) more runs than all the players that are part of a local team but scored (strictly) fewer runs than some designated hitter.

$$\exists x \forall y \exists z (S(x) \wedge (L(y) \rightarrow \neg P(x, y)) \wedge (D(z) \wedge P(x, z) \wedge \neg P(z, x)))$$

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For the given logical expression, choose the right English statement. It might not be a direct translation.

$$\forall x (S(x) \wedge D(x) \rightarrow \forall y (\neg M(y) \vee \neg L(y) \vee \neg P(x, y)))$$

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- ☐ Every designated hitter that is a starting pitcher scored as many or fewer runs than some player who is not on a local team or not an MVP.
- ☐ There exists a starting pitcher who is a designated hitter and scored more runs than every other player who is part of a local team or is an MVP.
- ☐ Every designated hitter that is a starting pitcher score as many or fewer runs than some player on a local team or an MVP.
- ☒ Every starting pitcher that is a designated hitter scored more runs than every player who is part of a local team and an MVP.